Xinhui Sun

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Education

University of Illinois at Urbana-Champaign — Champaign, IL
Ph.D. in Economics — Aug 2020 - May 2026 *(expected)*
Advisor: Prof. Tatyana Deryugina

University of Illinois at Urbana-Champaign — Champaign, IL
M.S. in Policy Economics — Dec 2019

University of International Business and Economics (UIBE) — Beijing, China
B.S. in Economics (Minor in French) — Jun 2018

University of California, Berkeley — Berkeley, CA
Exchange student — Jan 2017 - May 2017

Research Interests

Environmental Economics, Public Economics, Applied Econometrics

Working Papers

Air Pollution and Economic Activity: Evidence from Foot Traffic Patterns (Job Market Paper)
I investigate how air pollution affects economic activity. Using over 37 billion phone-location-based foot traffic data points from SafeGraph, I conduct a large-scale analysis to examine the causal effect of air pollution on activity patterns across the U.S. Using changes in local wind direction as an instrumental variable for air pollution, I characterize the dynamic response to pollution exposure. I find that PM2.5 reduces economic activity in both the short and medium run, with effects persisting for up to two weeks before gradually recovering. The reductions are widespread across different economic sectors, with recreational activities experiencing the largest decline. The effect is more pronounced in higher-income counties and areas with larger shares of children, suggesting greater awareness among wealthier or more vulnerable populations.

Go with the Wind: Polluters' Strategic Response to Wind Directions
I investigate the strategic behavior of polluters in the U.S. Under the Clean Air Act, air quality monitor data determine whether an area meets environmental standards. Because wind carries pollution, monitors detect more pollution from upwind sources and less from downwind sources. This disparity incentivizes polluters to emit less when they are upwind of a monitor and more when they are downwind. I identify such strategic behavior among U.S. coal-fired power plants, finding that a one-standard-deviation increase in favorable wind direction—when wind blows pollutants away from monitors—leads to a 0.8% (172 lbs) increase in sulfur dioxide (SO_2) emissions and a 0.4% (54 lbs) increase in nitrogen oxides (NO_x) emissions. At the same time, fuel input remains constant while emission rates rise, suggesting that power plants temporarily turn off pollution-control equipment. The increase is more pronounced when power plants are in a nonattainment county, located in the same state as the monitor, or surrounded by fewer nearby polluters. These findings suggest that polluters facing stricter regulatory pressure are more likely to respond strategically when conditions are favorable.

Work in Progress

Natural Disasters and Occupational Mobility: Evidence from the 1927 Mississippi Flood
(joint with Xiaohan Wang)

We investigate how a major natural disaster affects intergenerational occupational mobility in the agricultural sector. Linking individuals across the 1920, 1930, and 1940 Censuses, we show that children residing in counties affected by the 1927 Mississippi Flood were significantly less likely to become farmers as adults. This effect is mainly driven by children from non-farming families, who were 9% less likely to work in agriculture and 10% more likely to transition into the service sector. In the longer run, these individuals achieved socioeconomic status scores about 3% higher than comparable children in non-flooded counties. Our findings suggest that natural disasters can accelerate structural change by shifting new labor market entrants out of agriculture and into higher-status occupations.

Climate Change and Agricultural Adaptation in Peru
(joint with Tatyana Deryugina)

Social Networks and Household Protective Investments: Evidence from Wildfire Smoke
(joint with Sanjukta Mitra)

Deforestation and Air Pollution in the Indonesian Rainforest
(joint with Francesco Cenerini)

Conferences and Workshops

2026
ASSA Annual Meeting (Philadelphia)

2025
Canadian Resource and Environmental Economics Association Annual Conference (Winnipeg, Canada), Southern Economic Association Annual Conference (Tampa), Heartland Workshop (poster, UIUC), Program in Environmental and Resource Economics Seminar (UIUC), Midwest Economics Association Annual Conference (Kansas City), Eastern Economic Association Annual Conference (New York City), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

2024
Occasional Workshop in Environmental and Resource Economics (UCSB), Online Summer Workshop in Environment, Energy, and Transportation (Virtual), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

2023
Price Theory Summer Camp (University of Chicago), Applied Micro Research Lunch (UIUC)

2022
Berkeley/Solan Summer School in Environmental and Energy Economics, Applied Micro Research Lunch (UIUC)

Teaching Experience

ECON 102: Microeconomic Principles — Fall 2021, Spring 2022
Teaching Assistant

The Economics of the Firm (EMBA at the University of Warsaw) — Fall 2019
Teaching Assistant to Prof. Hadi Esfahani

ECON 528: Microeconomics for Business — Spring 2019, Summer 2019
Course Assistant

Work Experience

Gies College of Business, UIUC
Research Assistant to Prof. Tatyana Deryugina — Aug 2022 – present

Department of Economics, UIUC
Research Assistant to Prof. Eunyi Chung — June 2021 – Aug 2022

China Institute for Educational Finance Research, Peking University
Research Associate — Mar 2020 – Aug 2020

Risk Management Department, Bank of Communications
Data Analyst Intern — Jun 2017 – Aug 2017

School of International Trade and Economics, UIBE
Research Assistant to Prof. Deyu Yuan — Sep 2016 – Jan 2017

Awards

- Robert Willis Harbeson Memorial Dissertation Fellowship, University of Illinois — 2025
- Hung-Chao and Julia Tai Family Fellowship, University of Illinois — 2025
- Cleo Fitzsimmons Award, University of Illinois — 2022
- Graduate Fellowship, University of Illinois — 2020

Skills

Languages: Chinese (Native), English (Fluent), French (Basic).
Programming: `R`, `Python`, `Stata`, `Git`, `Shell`.

References

Tatyana Deryugina (Chair)
Associate Professor of Finance
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